

$$V_{th} = |V_{tp}| = 0.6 \text{ V}$$

$$\mu_n = 0.025 \text{ m}^2/\text{Vs} \quad \mu_p = 0.009 \text{ m}^2/\text{Vs}$$

$$C_{oxn} = 8.34 \text{ fF}/\mu\text{m}^2$$

$$C_{oxp} = 8.16 \text{ fF}/\mu\text{m}^2$$

$$V_{satn} = 8 \times 10^4 \text{ m/s}$$

$$V_{satp} = 3 \times 10^4 \text{ m/s}$$

$$V_{DD} = 1.8 \text{ V}$$

$$W_{min} = 0.42 \mu\text{m}$$

$$L_{min} = 0.15 \mu\text{m}$$

$$C_g = 0.8 \times \frac{W}{W_{min}} \text{ fF}$$

$$C_j = 0.16 \times \frac{W}{W_{min}} \text{ fF}$$

$$C_{jsw} = 0.2 \times \frac{W}{W_{min}} + 0.12 \text{ fF}$$

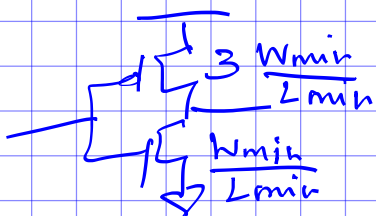
$$C_{gr} = 0.2 \times \frac{W}{W_{min}} \text{ fF}$$

1) 1) $W_p = ?$

$$V_{th} = 0.9 \text{ V}$$

nMOS & pMOS in SAT

$$W_p \approx 2.8 W_{min} = 3 W_{min}$$

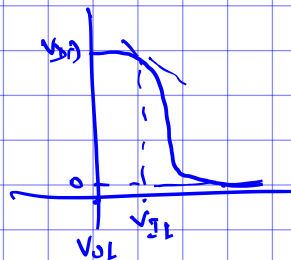


1.1.2.) NM_L

$nMOS = SAT$

$$V_{Dn} \gg V_{Gn}$$

$pMOS = \text{Linear}$



$$V_{in} = V_{IL}$$

$$\begin{array}{l} V_{GSn} = V_{in} \\ V_{DSn} = V_{out} \end{array} \left| \begin{array}{l} V_{SGp} = V_{DD} - V_{in} \\ V_{SDp} = V_{DD} - V_{out} \end{array} \right. \begin{array}{l} \frac{\partial V_{out}}{\partial V_{in}} = -1 \\ \rightarrow \approx 0 \end{array}$$

$$\begin{aligned} & \frac{\mu_n C_{ox} \left(\frac{W}{L}\right)_n}{2} (V_{in} - V_{Tn})^2 \frac{E_{cn} L}{E_{cn} L + V_{in} - V_{Tn}} \\ &= \mu_p C_{ox} \left(\frac{W}{L}\right)_p (2(V_{DD} - V_{in} - V_{Tp})(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2) \frac{E_{cp} L}{E_{cp} L + V_{DD} - V_{out}} \end{aligned}$$

$$\begin{aligned} K_n &= \frac{\mu_n C_{ox} \left(\frac{W}{L}\right)_n}{2} \\ K_p &= \frac{\mu_p C_{ox} \left(\frac{W}{L}\right)_p}{2} \end{aligned}$$

$$\begin{aligned} V_{in} &= \\ &\downarrow \\ &V_{IL} \end{aligned}$$

$$(2V_{out} - V_{DD} - |V_{Tp}|) K_p + K_n V_{Tn}$$

$$K_p + K_n$$

$$NM_L = V_{IL} = 0.908 V$$

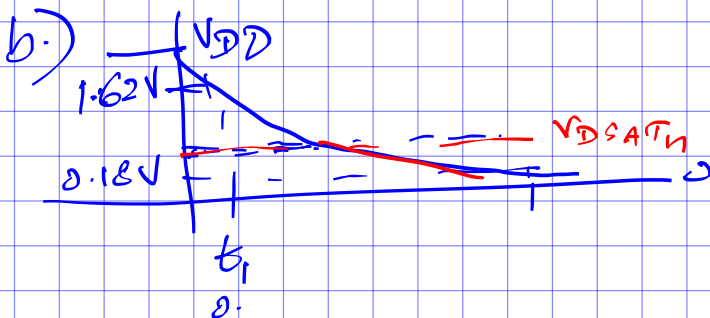
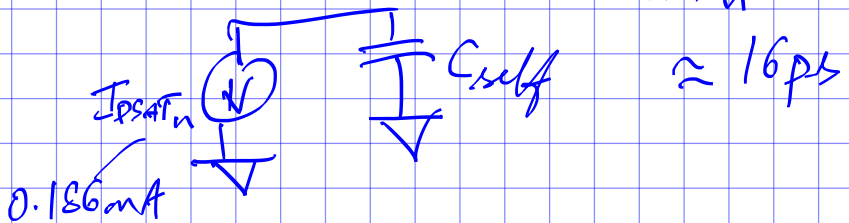
$$1.2.) C_{self} = \underset{\text{bottom}}{\text{junction}} + \underset{\text{sidewall}}{\text{junction}} + 2 \times \text{Overlap Cap}$$

$$= \underbrace{0.16 \times 4} + \underbrace{0.2 \times 4} + \underbrace{0.12 \times 2} + \underbrace{2 \times 0.2 \times 4}$$

$$= \underline{\underline{3.28 \text{ pF}}}$$

$$C_{inv} = 4 \times C_g = 4 \times 0.8 = \underline{\underline{3.2 \text{ pF}}}$$

$$1.2.2) a.) t_{PHL} = \frac{C_{self} \times V_{DD}}{2 I_{DSATn}}$$



$$V_{DSATn} = \frac{E_{cn} L x (V_{GS} - V_{tn})}{E_{cn} L + V_{GS} - V_{tn}}$$

$$V_{GS} \geq V_{tn}$$

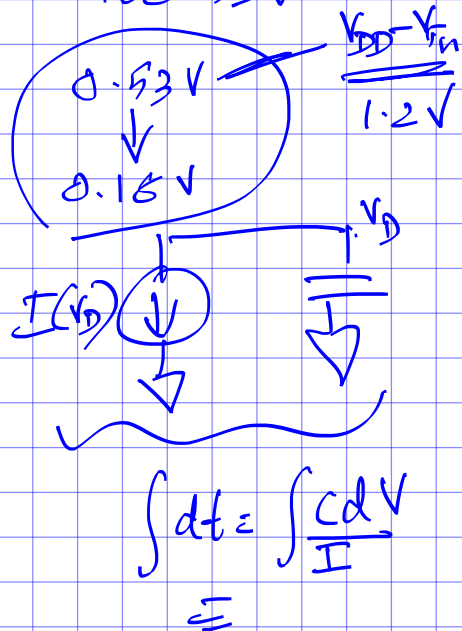
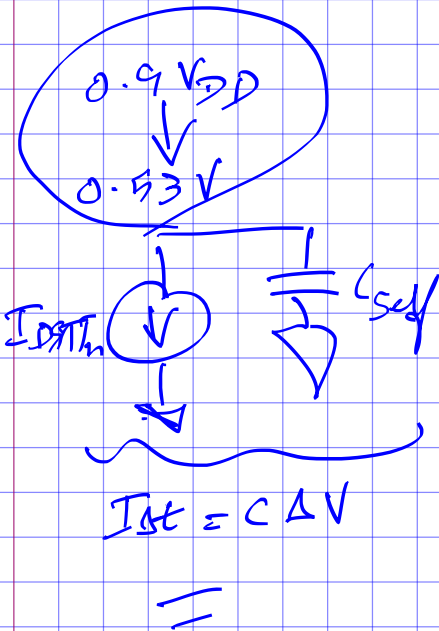
$$V_{DS} \geq V_{DSATn}$$

nMOS

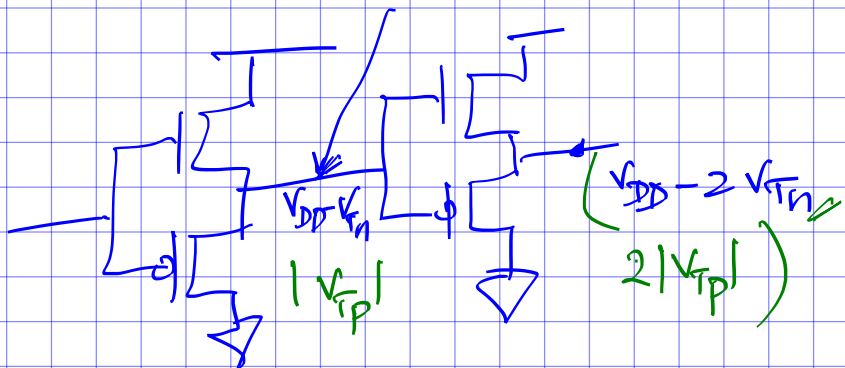
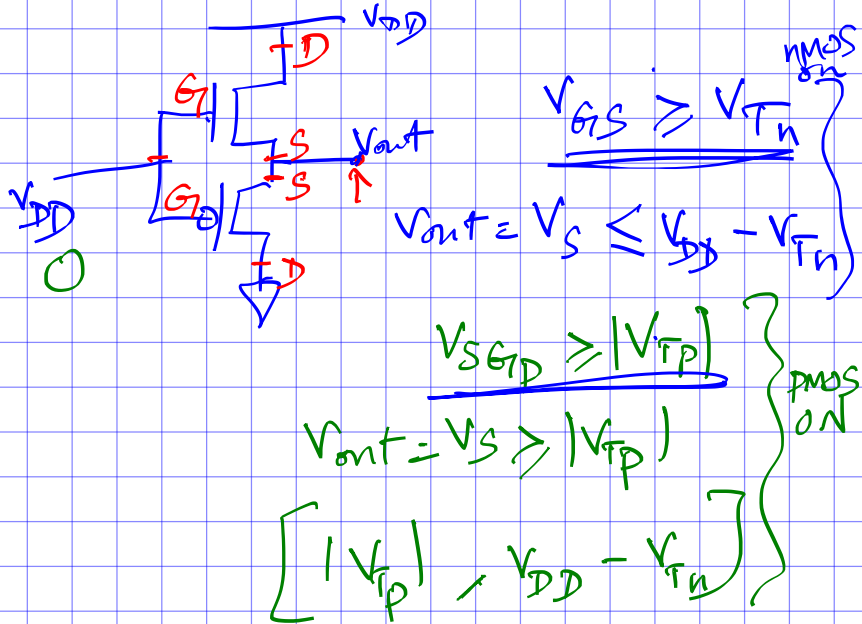
$$V_{DS} \geq V_{DSATn} = \text{SAT}$$

$$V_{DS} < V_{DSATn} = \text{Linear}$$

$$\sim 0.53V$$

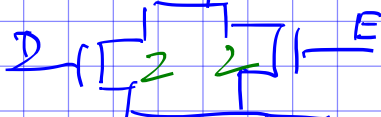


2.)

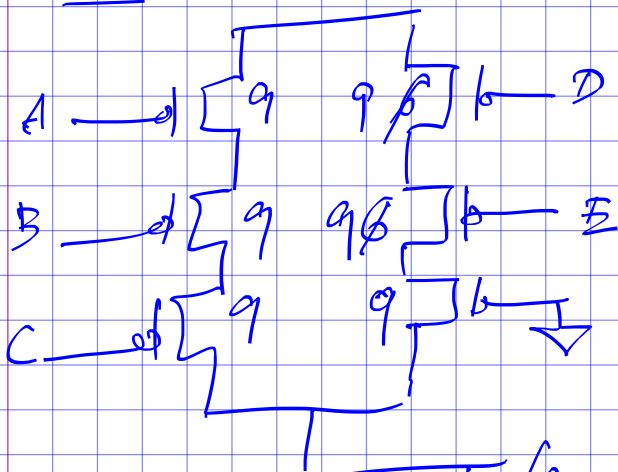


3.) $F = (A + B + C) \cdot (D + E)$

PDN



PUN

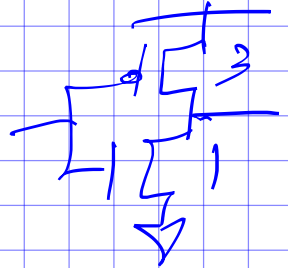


$$P = (9 + 9 + 2 + 2) C_g = 22 C_g$$

$$P = \frac{22}{4}$$

$$g = \frac{9 + 2}{4} = \frac{11}{4}$$

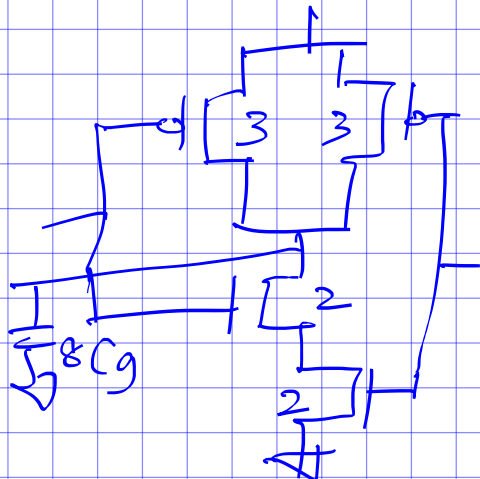
Ref inv

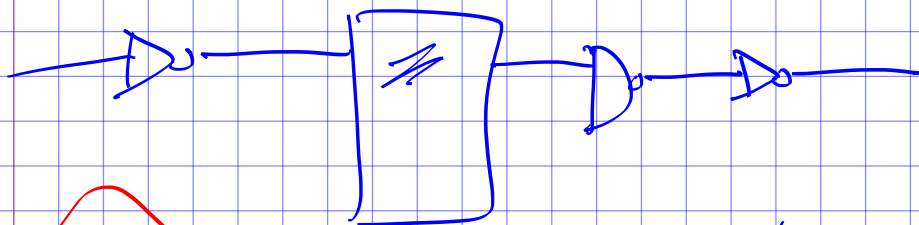


3.2)

$$P = \frac{8}{4} = 2$$

$$g = \frac{5}{4}$$





P
 g

$$\frac{11}{2}$$

$$2$$

$$1$$

$$\frac{11}{4}$$

$$\frac{5}{4}$$

$$1$$

$$F = G B H = \left(\frac{11}{4} \times \frac{5}{4} \right) \times 100$$

$$H = 100$$

$$g^* = F^{\frac{1}{4}} \approx 4.3$$

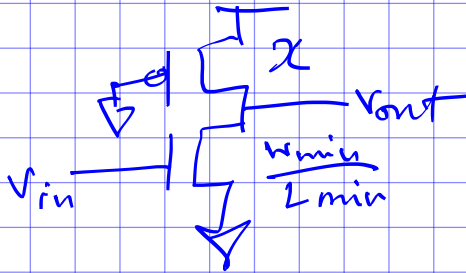
$$g_i h_i = g^*$$

$$\frac{1 \times \text{Cont}}{\text{Cinv}} = 4.3$$

$$3.2-3) \quad n = \log_4 F \quad 4 < n < 5$$

$$n^* = 5$$

4.) 1.)



$$V_{OL} = 0.1V$$

nMOS

$$V_{Gn} \gg V_{Dn}$$

Linear

PMOS

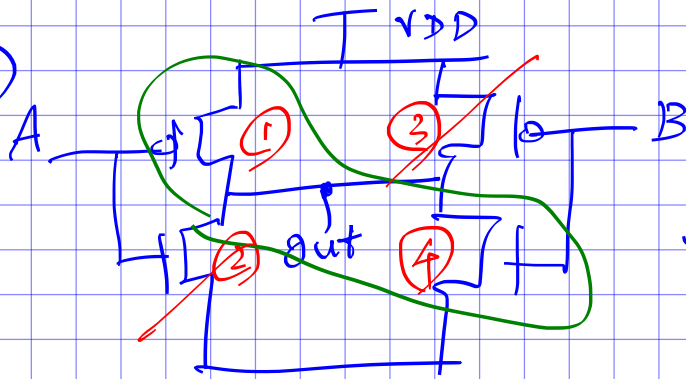
SAT

$$\frac{\mu_n C_{ox} \left(\frac{W}{L}\right)_n}{2} \left((V_{DD} - V_{Th}) V_{OL} - V_{OL}^2 \right) \frac{E_{cn} L}{E_{cn} L + V_{OL}}$$

$$= \frac{\mu_p C_{ox} \left(\frac{W}{L}\right)_p}{2} \left(V_{DD} - V_{Tp} \right)^2 \frac{E_{cp} L}{E_{cp} L + V_{DD} - V_{Tp}}$$

$$\left(\frac{W}{L}\right)_p \approx 0.45$$

4.2)



Johnson's
NOR
gate

A	B	out	
0	0	V_{DD}	Both p ON n OFF
0	V_{DD}	$V_{OL} = 0.1V$	① ON ④ ON
V_{DD}	0	$V_{OL} = 0.1V$	② & ③ ON
V_{DD}	V_{DD}	0	n ON p OFF

(NOR)